

Semi-blind Channel Estimation for Unsourced Random Access using Gram-guided Diffusion

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Abstract—We propose a semi-blind channel estimation framework for unsourced random access (URA) with multi-antenna receivers based on Gram-guided diffusion models. The method leverages a learned generative prior to model complex channel distributions and incorporates both pilot observations and an estimate of the channel Gram matrix derived from data symbols. This enables channel estimation that is jointly consistent with the prior as well as first-order and second-order statistics of the observations. The framework naturally handles non-orthogonal pilots and multi-user interference, addressing key challenges in practical URA systems. Simulation results show significant improvement over least-squares and sample-covariance based linear minimum-mean squared error estimators.

I. INTRODUCTION

Massive machine-type communication (mMTC) is a key service category in emerging wireless systems, characterized by a large number of devices transmitting short packets sporadically in a grant-free manner. Building on coded slotted ALOHA [1], [2], Polyanskiy’s information-theoretic formulation [3], and subsequent coding-theoretic advances [4], [5], Unsourced Random Access (URA) has become a central paradigm for modeling mMTC systems. In URA, all users share a common codebook and the receiver is tasked with recovering the transmitted messages without explicit user identification. Comprehensive treatments of URA can be found in [6]–[8].

While early work on URA primarily focused on the single-antenna additive white Gaussian noise (AWGN) channel, recent efforts have increasingly considered more realistic settings involving fading channels and multi-antenna receivers, which is the focus of this paper. In these scenarios, accurate channel estimation becomes a critical component of receiver design. Existing approaches include pilot-aided estimation, joint channel estimation and data detection, and iterative receiver architectures that combine linear minimum mean-squared error (LMMSE) estimation with soft interference cancellation. In particular, approximate message passing (AMP)-based techniques have been proposed for both channel estimation and joint detection [9]–[11], while alternative approaches exploit structured pilot designs and iterative refinement strategies [12]–[14].

Despite these advances, existing URA channel estimation methods exhibit two fundamental limitations. First, they use parametric priors such as joint Gaussianity, which is often

inaccurate in practice; estimators derived under these assumptions can suffer from model mismatch and poor performance. Second, existing methods do not fully exploit the structural information contained in the data portion of the received signal. In particular, second-order statistics such as the channel Gram matrix, obtainable from data symbols in a semi-blind manner, are not systematically integrated into modern learning-based or message-passing receivers. These challenges are exacerbated in URA, because pilots associated with different users are typically non-orthogonal, having been drawn pseudo-randomly from a shared pool, motivating a semi-blind approach that exploits the Gram matrix alongside non-orthogonal pilots, as developed in this paper.

We propose a diffusion-based semi-blind channel estimation framework for URA that addresses both limitations identified above. Rather than relying on parametric priors, we adopt a data-driven generative prior learned via diffusion models, which captures complex, realistic channel distributions. While a few recent papers have used diffusion pipelines for single-user MIMO channel estimation [15]–[17], we are unaware of papers that use diffusion pipelines for channel estimation for unsourced random access. Building on our Gram-guided diffusion framework [18], we formulate channel estimation as a conditional generation problem in which a pretrained diffusion model is guided by (i) pilot observations and (ii) an estimate of the channel Gram matrix obtained from the data portion of the received signal. The Gram matrix provides a realization-specific, second-order structural constraint that complements the pilots [19]. We extend the Gram-guided diffusion framework in [18] to the URA setting, where non-orthogonal pilots and multi-user superposition require new guidance mechanisms. By integrating these two sources of information within the diffusion inference process, the proposed method effectively performs semi-blind channel estimation in a principled and flexible manner.

We summarize our main contributions as follows:

- We introduce a diffusion-based framework for semi-blind channel estimation in URA, enabling data-driven inference without reliance on parametric channel models.
- We extend Gram-guided diffusion to the URA setting, addressing the challenges of non-orthogonal pilots.
- We show through simulations that the proposed method

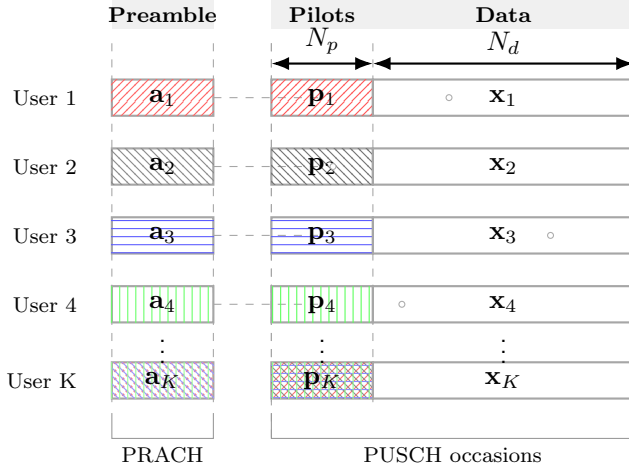


Fig. 1: Schematic of the unsourced random access scheme based on the 2-step RACH.

achieves significant gains over the baselines of Section IV, including under non-orthogonal pilots and at high SNR.

II. SYSTEM MODEL AND NOTATION

We consider a synchronous unsourced random access uplink with K single-antenna users and an N_R -antenna base station, where each user employs a grant-free random access protocol similar to the 2-step RACH procedure [20]. In this protocol (Fig. 1), user k selects a preamble $\mathbf{a}_k \in \mathcal{A}$ from a preamble set $\mathcal{A}$ and transmits it in the Physical Random Access Channel (PRACH) slot during step 1. In the subsequent transmission over the Physical Uplink Shared Channel (PUSCH), the user sends a pilot sequence $\mathbf{p}_k \in \mathbb{C}^{1 \times N_p}$, chosen from the pilot set $\mathcal{P}$, followed by modulated and encoded data $\mathbf{x}_k \in \mathbb{C}^{1 \times N_d}$. The pilot $\mathbf{p}_k$ is determined by the selected preamble $\mathbf{a}_k$.

We assume that the channel coherence time is longer than the frame length; hence, the fading is quasi-static. At time i , the received pilot or data signals are given by

$$\mathbf{y}_i = \sum_k \mathbf{h}_k p_{k,i} + \mathbf{w}_i, \quad i = 1, \dots, N_p \quad (1)$$

$$\mathbf{y}_i = \sum_k \mathbf{h}_k x_{k,i} + \mathbf{w}_i, \quad i = N_p + 1, \dots, N_p + N_d \quad (2)$$

where $p_{k,i}$ and $x_{k,i}$ denote the i th pilot symbol and the i th data symbol of user k , and $\mathbf{w}_i \sim \mathcal{CN}(0, \sigma^2 I_{N_R})$ is a vector of independent, zero-mean Gaussian random components. The channel vector $\mathbf{h}_k \in \mathbb{C}^{N_R}$ is unknown to both transmitter and receiver, and is the quantity to be estimated. As can be seen from (1) and (2), pilots are transmitted during the first N_p time instants and data is transmitted during the next N_d time instants.

Collecting these samples over time yields matrices $\mathbf{Y}_p \in \mathbb{C}^{N_R \times N_p}$ and $\mathbf{Y}_d \in \mathbb{C}^{N_R \times N_d}$. Let $\mathbf{P} \in \mathbb{C}^{K \times N_p}$ and $\mathbf{X} \in \mathbb{C}^{K \times N_d}$ denote the pilot and data matrix where the k th row of $\mathbf{P}$ and $\mathbf{X}$ denote the pilot and data symbols, respectively,

of the k th user. Let $\mathbf{H} \in \mathbb{C}^{N_R \times K}$ denote the channel matrix whose k th column is $\mathbf{h}_k$. Then

$$\mathbf{Y}_p = \mathbf{H}\mathbf{P} + \mathbf{W}_p, \quad (3)$$

$$\mathbf{Y}_d = \mathbf{H}\mathbf{X} + \mathbf{W}_d. \quad (4)$$

At the base station, active preambles are detected first, and $\mathbf{P}$ is thereby identified. The primary task under consideration is estimating $\mathbf{H}$ from $\mathbf{P}$, $\mathbf{Y}_p$, and $\mathbf{Y}_d$.

Unlike much of the existing literature, we do not assume a parametric model for $\mathbf{H}$. Instead, we leverage a dataset of channel realizations and develop a diffusion-based pipeline for channel estimation. Because the pilot assignments are random and depend on the random access preambles, the columns of $\mathbf{P}$ are typically not orthogonal.

We assume that the rows of $\mathbf{P}$ are each normalized to have unit energy for all N_p and, therefore, $\text{diag}(\mathbf{P}\mathbf{P}^H) = I_{N_R}$. The pilot signal to noise ratio is then defined as $\text{SNR} := 1/\sigma^2$.

For any $N_R \times K$ matrix $\mathbf{H}$, let $\tilde{\mathbf{H}}$ denote the two-dimensional Discrete Fourier Transform (DFT) of $\mathbf{H}$ given by

$$\tilde{\mathbf{H}} = \Phi_{N_R} \mathbf{H} \Phi_K^T, \quad (5)$$

where Φ_N is the 1-D DFT matrix of size $N \times N$ representing a N -point DFT. For a uniform linear array, $\tilde{\mathbf{H}}$ represents the channel in the angular domain. It is well known that typical wireless channels have (sparse) representations in the angular domain, and [15] showed that simple neural networks suffice to learn priors in the angular domain through a diffusion process. We follow their approach and work with the angular representation in this paper as well.

Let $\mathbf{R} := \mathbf{H}\mathbf{H}^H$ and $\tilde{\mathbf{R}} := \tilde{\mathbf{H}}\tilde{\mathbf{H}}^H$ denote the Gram matrices of the rows of $\mathbf{H}$ and $\tilde{\mathbf{H}}$, respectively. The two Gram matrices are related through a similarity transformation given by $\tilde{\mathbf{R}} = \Phi_{N_R} \mathbf{R} \Phi_{N_R}^H$.

Note: Throughout the rest of the paper, capital boldface letters represent matrices in the spatial domain and the corresponding variables with tilde refer to matrices in the angular domain. Superscripts H , T , and $*$ denote the Hermitian, transpose, and conjugate of a matrix, respectively.

III. GRAM-GUIDED SEMI-BLIND CHANNEL ESTIMATION ALGORITHM

Our proposed algorithm is a diffusion-based channel estimation framework that leverages a pretrained generative prior for the angular-domain channel matrix, together with guidance derived from the received pilot and data observations. The diffusion model is trained offline to generate samples of $\tilde{\mathbf{H}}$, while during inference the observations $\mathbf{Y}_p$ and $\mathbf{Y}_d$ guide the reverse diffusion process toward channel realizations consistent with the uplink measurements.

A. Training

Our training starts with a dataset of channel matrices $\mathcal{H} = \{\mathbf{H}^{(m)}\}_{m=1}^M$, where each $\mathbf{H}^{(m)} \in \mathbb{C}^{N_R \times K}$. We derive this dataset from the 3GPP dataset $\mathcal{B}$ in [15], which contains 16×64 channel matrices representing a single-user MIMO setup

with 16 transmit and 64 receive antennas. We construct $\mathcal{H}$ by randomly sampling columns from the channel matrices in $\mathcal{B}$ to represent the random access scenario where each UE is equipped with a single transmit antenna. As in [15], the dataset is transformed to the angular domain via a 2-D DFT, yielding $\tilde{\mathcal{H}} = \{\tilde{\mathbf{H}}^{(m)}\}_{m=1}^M$.

A diffusion model is then trained on $\tilde{\mathcal{H}}$ to learn a generative prior for the angular-domain channel. During inference, all computations are performed in the angular domain, and the final estimate is converted back to the spatial domain. At step t of the reverse process, the learned denoiser $\mathcal{D}_t$ is applied to the estimate $\tilde{\mathbf{H}}_t$ to obtain $\tilde{\mathbf{H}}_{t-1} = \mathcal{D}_t(\tilde{\mathbf{H}}_t)$, where

$$\mathcal{D}_t(\tilde{\mathbf{H}}_t) = \sqrt{\bar{\alpha}_{t-1}} \hat{\tilde{\mathbf{H}}}_0(\tilde{\mathbf{H}}_t, t) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\tilde{\mathbf{H}}_t, t), \quad (6)$$

where $\epsilon_\theta(\tilde{\mathbf{H}}_t, t)$ is the learned noise predictor, and the estimate of clean signal is given by

$$\hat{\tilde{\mathbf{H}}}_0(\tilde{\mathbf{H}}_t, t) = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\tilde{\mathbf{H}}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\tilde{\mathbf{H}}_t, t) \right). \quad (7)$$

B. Preprocessing and Worst-Case Spectral Matching

Consider the received signal corresponding to the pilots in (3). To decouple the per-user channels, we right-multiply both sides of (3) by $\mathbf{P}^H(\mathbf{P}\mathbf{P}^H)^{-1}$ and obtain

$$\mathbf{Y}' \triangleq \mathbf{Y}_p \mathbf{P}^H(\mathbf{P}\mathbf{P}^H)^{-1} = \mathbf{H} + \mathbf{W}', \quad (8)$$

where

$$\mathbf{W}' \triangleq \mathbf{W}_p \mathbf{P}^H(\mathbf{P}\mathbf{P}^H)^{-1}. \quad (9)$$

Since the columns of $\mathbf{P}$ are not orthogonal, $\mathbf{W}'$ is correlated across users. Indeed, for each row vector $\mathbf{w}'_r$ of $\mathbf{W}'$, we have

$$\mathbf{w}'_r = \mathbf{w}_{p,r} \mathbf{P}^H(\mathbf{P}\mathbf{P}^H)^{-1}, \quad (10)$$

and

$$\mathbf{w}'_r \sim \mathcal{CN}(\mathbf{0}, \sigma^2(\mathbf{P}\mathbf{P}^H)^{-1}). \quad (11)$$

We then transform the model into the angular domain via a 2-D DFT:

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{H}} + \tilde{\mathbf{W}}, \quad (12)$$

where $\tilde{\mathbf{W}}$ remains colored with covariance given by the unitary transformation of $\sigma^2(\mathbf{P}\mathbf{P}^H)^{-1}$.

We adopt an SNR matching strategy to initialize the reverse diffusion process at an appropriate timestep. By aligning the diffusion noise level with that of the observation, the number of reverse steps is reduced, avoiding over-denoising and improving estimation accuracy.

Assuming $\mathbb{E}[\|\mathbf{H}\|_F^2] = N_R K$ and noting that the preprocessed noise has row covariance $\sigma^2(\mathbf{P}\mathbf{P}^H)^{-1}$, we adopt a worst-case spectral characterization and define

$$\sigma_{\text{eff}}^2 \triangleq \lambda_{\max}(\sigma^2(\mathbf{P}\mathbf{P}^H)^{-1}) = \frac{\sigma^2}{\lambda_{\min}(\mathbf{P}\mathbf{P}^H)}, \quad (13)$$

leading to $\text{SNR}_{\text{eff}} = \frac{1}{\sigma_{\text{eff}}^2}$. We employ a variance-preserving diffusion model, where each timestep corresponds to a specific noise level

$$\text{SNR}_{\text{DM}}(t) = \frac{\bar{\alpha}_t}{1 - \bar{\alpha}_t}. \quad (14)$$

We initialize the reverse diffusion process at the timestep whose SNR best matches the observation:

$$t^* = \arg \min_t |\text{SNR}_{\text{eff}} - \text{SNR}_{\text{DM}}(t)|. \quad (15)$$

The proposed SNR_{eff} captures the noise level along the most adverse direction induced by the colored covariance, providing a conservative scalar approximation for SNR matching. This worst-case characterization enables a robust and effective initialization of the reverse diffusion process, particularly under non-orthogonal pilot designs where the noise is highly anisotropic. Note that $\tilde{\mathbf{W}}$ is colored, while the diffusion forward process is defined with isotropic Gaussian noise. Hence, no exact timestep correspondence exists between the two noise models. Nevertheless, as shown in the next section, the covariance of the noise is accounted for when computing the likelihood in the guidance.

C. Gram-guided Diffusion

Starting from (8), where each row of $\mathbf{W}'$ satisfies $\mathbf{w}'_r \sim \mathcal{CN}(\mathbf{0}, \sigma^2(\mathbf{P}\mathbf{P}^H)^{-1})$, the resulting likelihood is

$$\log p(\mathbf{Y}' | \mathbf{H}) = \text{const} - \frac{1}{\sigma^2} \text{tr}[(\mathbf{Y}' - \mathbf{H})(\mathbf{P}\mathbf{P}^H)(\mathbf{Y}' - \mathbf{H})^H], \quad (16)$$

with gradient

$$\nabla_{\mathbf{H}} \log p(\mathbf{Y}' | \mathbf{H}) = \frac{1}{\sigma^2} (\mathbf{Y}' - \mathbf{H})(\mathbf{P}\mathbf{P}^H). \quad (17)$$

The likelihood guidance in the angular domain is given by

$$g_{\text{like}}(\tilde{\mathbf{H}}_t) = \frac{1}{\sigma^2} \left(\tilde{\mathbf{Y}} - \hat{\tilde{\mathbf{H}}}_0(\tilde{\mathbf{H}}_t, t) \right) (\Phi_K^* \mathbf{P} \mathbf{P}^H \Phi_K^T). \quad (18)$$

Following [18], we also exploit second-order structure via the channel Gram matrix. From the data observations in (4), we form the sample covariance

$$\frac{1}{N_d} \mathbf{Y}_d \mathbf{Y}_d^H \approx \mathbf{H} \mathbf{H}^H + \sigma_d^2 \mathbf{I}, \quad (19)$$

and obtain the Gram estimate

$$\hat{\mathbf{R}} = \frac{1}{N_d} \mathbf{Y}_d \mathbf{Y}_d^H - \sigma_d^2 \mathbf{I}. \quad (20)$$

Transforming to the angular domain gives $\hat{\mathbf{R}} = \Phi_{N_R} \hat{\mathbf{R}} \Phi_{N_R}^H$. We define the Gram consistency objective

$$f(\tilde{\mathbf{H}}_t) = - \left\| \tilde{\mathbf{H}}_t \tilde{\mathbf{H}}_t^H - \hat{\mathbf{R}} \right\|_F^2, \quad (21)$$

whose gradient yields

$$\mathbf{g}_{\text{gram}}(\tilde{\mathbf{H}}_t) = 4(\hat{\mathbf{R}} - \tilde{\mathbf{H}}_t \tilde{\mathbf{H}}_t^H) \tilde{\mathbf{H}}_t. \quad (22)$$

Combining with the likelihood guidance derived previously, the reverse update is

$$\tilde{\mathbf{H}}_{t-1} = \mathcal{D}_t(\tilde{\mathbf{H}}_t) + \eta_{\text{like},t} \mathbf{g}_{\text{like}}(\tilde{\mathbf{H}}_t) + \eta_{\text{gram},t} \mathbf{g}_{\text{gram}}(\tilde{\mathbf{H}}_t). \quad (23)$$

To match the diffusion dynamics, we adopt

$$\eta_{\text{like},t} = \beta_t w_{\text{like}}(\text{SNR}), \quad (24)$$

$$\eta_{\text{gram},t} = \sqrt{\beta_t} w_{\text{gram}}(\text{SNR}), \quad (25)$$

where $w_{\text{like}}(\cdot)$ and $w_{\text{gram}}(\cdot)$ are SNR-dependent scheduling functions. The overall algorithm is given below.

Algorithm 1 Gram-Guided Diffusion

Require: Pilot observation $\mathbf{Y}_p$, data observation $\mathbf{Y}_d$, pilot matrix $\mathbf{P}$, pretrained diffusion denoiser $\{\mathcal{D}_t\}_{t=1}^T$

Ensure: Channel estimate $\hat{\mathbf{H}}$

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1:  $\tilde{\mathbf{Y}} \leftarrow \Phi_{N_R} \mathbf{Y}_p \mathbf{P}^H (\mathbf{P} \mathbf{P}^H)^{-1} \Phi_K^T$ 
2:  $\text{SNR}_{\text{eff}} \leftarrow \lambda_{\min}(\mathbf{P} \mathbf{P}^H) / \sigma^2$ 
3:  $t^* \leftarrow \arg \min_t |\text{SNR}_{\text{eff}} - \text{SNR}_{\text{DM}}(t)|$ 
4:  $\tilde{\mathbf{H}}_{t^*} \leftarrow \tilde{\mathbf{Y}} / \sqrt{1 + 1/\text{SNR}_{\text{eff}}}$ 
5:  $\hat{\mathbf{R}} \leftarrow \frac{1}{N_d} \mathbf{Y}_d \mathbf{Y}_d^H - \sigma_d^2 \mathbf{I}$ 
6:  $\hat{\mathbf{R}} \leftarrow \Phi_{N_R} \hat{\mathbf{R}} \Phi_{N_R}^H$ 
7: for  $t = t^*, t^* - 1, \dots, 1$  do
8:    $\hat{\mathbf{H}}_0 \leftarrow \frac{1}{\sqrt{\alpha_t}} \left( \tilde{\mathbf{H}}_t - \sqrt{1 - \alpha_t} \epsilon_\theta(\tilde{\mathbf{H}}_t, t) \right)$ 
9:    $g_{\text{like}} \leftarrow \frac{1}{\sigma^2} \left( \tilde{\mathbf{Y}} - \hat{\mathbf{H}}_0 \right) \left( \Phi_K^* \mathbf{P} \mathbf{P}^H \Phi_K^T \right)$ 
10:   $\mathbf{g}_{\text{gram}} \leftarrow 4 \left( \hat{\mathbf{R}} - \tilde{\mathbf{H}}_t \tilde{\mathbf{H}}_t^H \right) \tilde{\mathbf{H}}_t$ 
11:   $\hat{\mathbf{H}}_{t-1} \leftarrow \mathcal{D}_t(\tilde{\mathbf{H}}_t) + \eta_{\text{like},t} g_{\text{like}} + \eta_{\text{gram},t} \mathbf{g}_{\text{gram}}$ 
12: end for
13:  $\hat{\mathbf{H}} \leftarrow \Phi_{N_R}^H \hat{\mathbf{H}}_0 \Phi_K^*$ 

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IV. BASELINES

In this section, we review some baseline estimation algorithms used for comparing the performance of the proposed algorithm. For any matrix $\mathbf{X}$, we define $\text{vec}(\mathbf{X})$ as the operator that vectorizes the matrix columnwise. Vectorizing (3) yields

$$\mathbf{y}_p = \text{vec}(\mathbf{Y}_p) = (\mathbf{P}^T \otimes \mathbf{I}_{N_R}) \mathbf{h} + \mathbf{n} = \mathbf{A} \mathbf{h} + \mathbf{w}_p, \quad (26)$$

where $\mathbf{h} = \text{vec}(\mathbf{H})$, $\mathbf{w}_p = \text{vec}(\mathbf{W}_p)$, and $\mathbf{A} = \mathbf{P}^T \otimes \mathbf{I}_{N_R}$.

We present results for the 3GPP dataset considered in [15]. In this dataset, the channel matrices are drawn from a mixture Gaussian model with a latent vector δ , which represents the angle of departures and angle of arrivals. Conditioned on δ , $\mathbf{h} = \text{vec}(\mathbf{H}) \mid \delta \sim \mathcal{CN}(0, \mathbf{C}_\delta)$.

For the random access setting considered here, the transmit-side covariance reduces to identity, so that $\mathbf{C}_\delta = \mathbf{I}_K \otimes \mathbf{C}_{rx,\delta}$, which implies that the columns of $\mathbf{H}$ are conditionally independent and hence, $\mathbf{h}_k \mid \delta \sim \mathcal{CN}(0, \mathbf{C}_{rx,\delta})$, for $k = 1, \dots, K$.

A. Least Squares (LS)

The LS estimator is given by

$$\hat{\mathbf{H}}_{\text{LS}} = \mathbf{Y}_p \mathbf{P}^H (\mathbf{P} \mathbf{P}^H)^{-1}. \quad (27)$$

B. Genie-aided LMMSE

Assuming δ is known, the conditional MMSE estimator is

$$\hat{\mathbf{h}}_{\text{G-LMMSE}} = \mathbf{C}_\delta \mathbf{A}^H (\mathbf{A} \mathbf{C}_\delta \mathbf{A}^H + \sigma^2 \mathbf{I})^{-1} \mathbf{y}_p. \quad (28)$$

C. Sample Covariance LMMSE (Scov-LMMSE)

In practice, $\mathbf{C}_\delta$ is unknown and is replaced by the sample covariance

$$\hat{\mathbf{C}} = \frac{1}{M} \sum_{m=1}^M \mathbf{h}^{(m)} (\mathbf{h}^{(m)})^H. \quad (29)$$

The resulting estimator is

$$\hat{\mathbf{h}}_{\text{scov}} = \hat{\mathbf{C}} \mathbf{A}^H (\mathbf{A} \hat{\mathbf{C}} \mathbf{A}^H + \sigma^2 \mathbf{I})^{-1} \mathbf{y}_p. \quad (30)$$

D. Genie-aided LMMSE

We now consider a genie-aided estimator that is given the Gram matrix $\mathbf{R}$ and the latent vector δ . Since $\mathbf{R} = \mathbf{H} \mathbf{H}^H = \mathbf{U} \Sigma^2 \mathbf{U}^H$, knowing $\mathbf{R}$ reveals the column subspace of $\mathbf{H}$. Hence,

$$\mathbf{H} = \mathbf{U} \mathbf{S}, \quad (31)$$

where $\mathbf{U} \in \mathbb{C}^{N_R \times r}$ and $\mathbf{S} \in \mathbb{C}^{r \times K}$.

Projecting the pilot observation onto this subspace gives

$$\tilde{\mathbf{Y}} = \mathbf{U}^H \mathbf{Y}_p = \mathbf{S} \mathbf{P} + \tilde{\mathbf{N}}. \quad (32)$$

Equivalently, in vectorized form,

$$\tilde{\mathbf{y}} = \text{vec}(\tilde{\mathbf{Y}}) = \mathbf{A}_r \mathbf{s} + \tilde{\mathbf{n}}, \quad (33)$$

where $\mathbf{A}_r = \mathbf{P}^T \otimes \mathbf{I}_r$. Since $\mathbf{h}_k = \mathbf{U} \mathbf{s}_k$, we have $\mathbf{s}_k \mid \delta \sim \mathcal{CN}(0, \mathbf{U}^H \mathbf{C}_{rx,\delta} \mathbf{U})$. Stacking all columns yields

$$\mathbf{s} \mid \delta \sim \mathcal{CN}(0, \Sigma_{S,\delta}), \quad (34)$$

with $\Sigma_{S,\delta} = \mathbf{I}_K \otimes (\mathbf{U}^H \mathbf{C}_{rx,\delta} \mathbf{U})$. Therefore, the LMMSE estimator in the reduced subspace is

$$\hat{\mathbf{s}} = \Sigma_{S,\delta} \mathbf{A}_r^H (\mathbf{A}_r \Sigma_{S,\delta} \mathbf{A}_r^H + \sigma^2 \mathbf{I})^{-1} \tilde{\mathbf{y}}. \quad (35)$$

Let $\hat{\mathbf{S}} = \text{unvec}(\hat{\mathbf{s}})$. Note that this LMMSE estimator does not explicitly enforce the Gram constraint $\hat{\mathbf{S}} \hat{\mathbf{S}}^H = \Sigma^2$, which is known from $\mathbf{R}$. To incorporate this constraint, we perform a whitening and recoloring operation to obtain $\mathbf{S}_{\text{proj}}$ given by

$$\mathbf{S}_{\text{proj}} = \Sigma (\hat{\mathbf{S}} \hat{\mathbf{S}}^H + \kappa \mathbf{I})^{-1/2} \hat{\mathbf{S}}, \quad (36)$$

where $\kappa > 0$ is a small regularization parameter for numerical stability. When $\kappa = 0$, it can be seen that $\mathbf{S}_{\text{proj}} \mathbf{S}_{\text{proj}}^H = \Sigma^2$. The final channel estimate is then given by

$$\hat{\mathbf{H}}_{\text{Genie-LMMSE-Gram}} = \mathbf{U} \mathbf{S}_{\text{proj}}. \quad (37)$$

V. SIMULATION RESULTS

We consider the unsourced random access setting described in Section II with $K = 16$, $N_R = 64$, and $N_p = 16, 32$. The pilot set $\mathcal{P}$ is constructed by choosing the pilot sequences to have i.i.d. complex Gaussian entries $\mathcal{CN}(0, \frac{1}{N_p})$. Every pilot sequence is then normalized to have unit energy. Under this normalization, the pilot matrix chosen by the active users satisfies $\text{diag}(\mathbf{P} \mathbf{P}^H) = \mathbf{I}_K$. We assume that each user transmits 100 bits of information encoded into a low rate code of rate-1/20 resulting in $N_d = 2000$.

We present results for the 3GPP dataset in [15]. As mentioned in Section III-A, we modify this dataset to reflect the random access scenario and the diffusion model is trained in the angular domain.

A plot of the normalized mean-squared error versus the pilot signal-to-noise ratio is shown in Fig. 2. It can be seen from the plot that the performance of the diffusion model with likelihood guidance is more than 5 dB better than that of the sample covariance based LMMSE estimator and is significantly better than that of the least-squares estimator. Adding GRAM-based guidance further improves the performance significantly. Finally, it can be noted that the performance is only a few

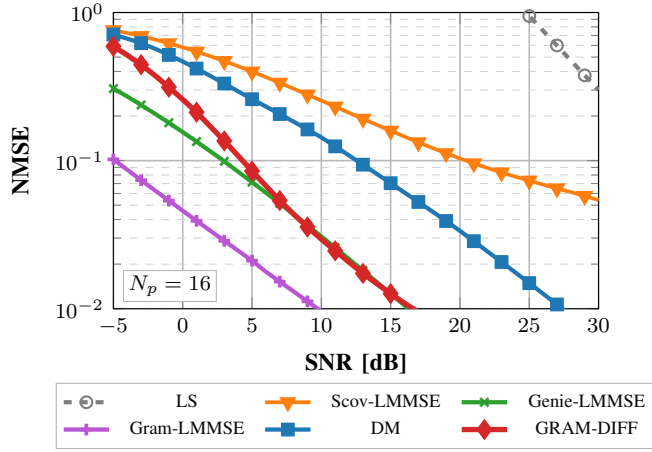


Fig. 2: Plot of normalized mean-squared error versus signal-to-noise ratio; $K = 16$, $N_R = 64$, $N_p = 16$, $N_d = 2000$.

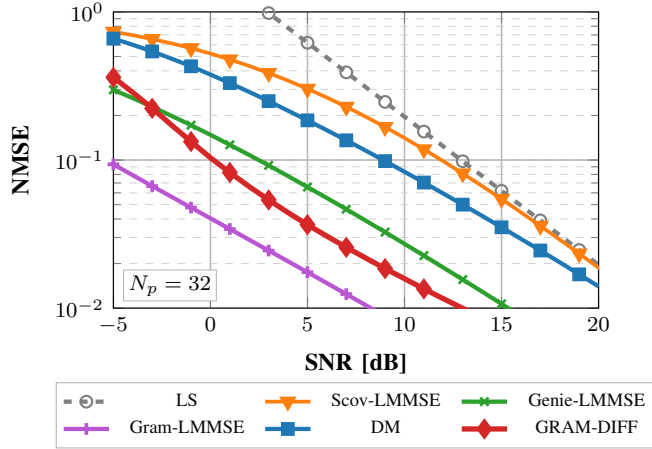


Fig. 3: Plot of normalized mean-squared error versus signal-to-noise ratio; $K = 16$, $N_R = 64$, $N_p = 32$, $N_d = 2000$.

dB away from the genie-aided LMMSE with Gram guided estimator. Another set of results are shown in Fig. 3 when $N_p = 32$. The diffusion model still provides improvement over the sample covariance-based estimator and least-squares, although the gap is now decreased for high SNRs. With Gram guidance, the proposed technique even outperforms the genie-aided linear MMSE estimator owing to the realization level information in the Gram matrix.

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